

AI-Driven Multimodal Emotional Recognition System with Explainability

Group Name: MedOracle • Group Number: 36

Problem in Brief

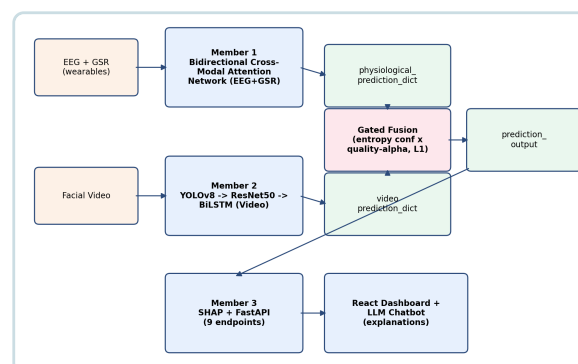
- Mental health assessment still relies mainly on subjective self-report; continuous, objective measurement is limited.
- Single-modality tools (a camera OR a wearable, never both) inherit that one channel's blind spots — a face can be controlled, physiology cannot, but physiology varies heavily between people.
- Existing multimodal systems assume every input is always present and equally reliable, and fuse with fixed weights that cannot adapt when a sensor is degraded or missing.
- The two best available datasets for physiological (DEAP) and facial-video (CREMA-D) emotion recognition share no common subjects — genuine multimodal fusion cannot be trained or evaluated the usual way.

Aim & Objectives

Aim: To design, implement and rigorously evaluate a multimodal emotion recognition system that fuses physiological (EEG/GSR) and facial-video signals through a quality-aware gated fusion mechanism, and to make its predictions explainable through SHAP.

Objectives

- Physiological emotion recognition via a bidirectional cross-modal attention network (DEAP).
- Video emotion recognition via ResNet50 + BiLSTM, evaluated actor-independently (CREMA-D).
- A quality-aware gated fusion mechanism with graceful degradation, verified by automated testing.
- SHAP-based explainability + a FastAPI/React web app accepting real multimodal input.
- An 8-condition ablation study with statistical significance testing — reporting all results honestly, including unfavourable ones.

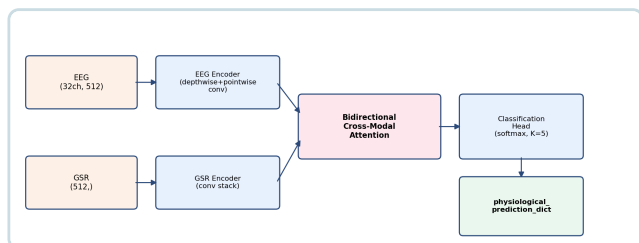


System architecture — physiological and video branches fused via quality-aware gating, explained via SHAP

MODULE 01 Physiological Emotion Recognition

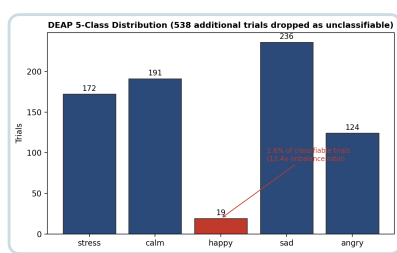
Bidirectional cross-modal attention network over EEG (32-channel) and GSR, trained on DEAP, evaluated honestly under Leave-One-Subject-Out cross-validation.

METHODOLOGY



Module 01 pipeline — EEG/GSR preprocessing to bidirectional attention classifier

EVALUATION & RESULTS



DEAP class distribution — severe imbalance, only 2.6% happy

LOSO macro-F1 (honest, subject-independent)

0.179

Random-split macro-F1 (optimistic baseline)

0.417

- Root cause diagnosed via 4 controlled experiments — traced to class imbalance and cross-subject EEG variability, NOT a feature-engineering or implementation defect.

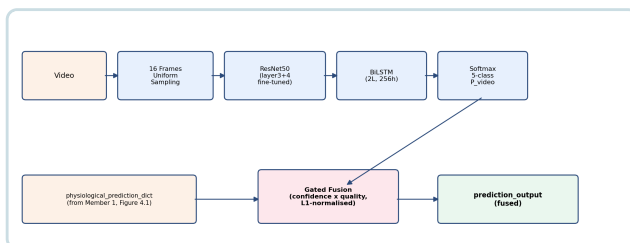
KEY FEATURES

- Bidirectional cross-modal attention (EEG ↔ GSR).
- Signal-quality grading (good/degraded/poor) feeding directly into system-level fusion.
- Honest dual-protocol evaluation (random-split vs LOSO).

MODULE 02 Video Recognition + Gated Fusion

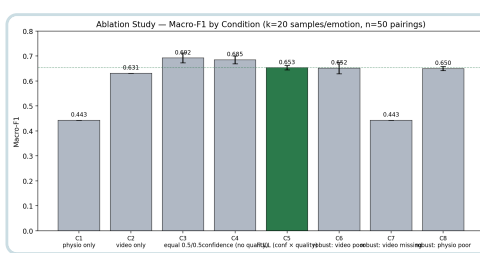
YOLOv8 face detection (quality grading only) → partially fine-tuned ResNet50 → BiLSTM → 5-class softmax, fused with Module 01 via an adaptive, confidence-and-quality-aware gate.

METHODOLOGY



Module 02 pipeline — video recognition branch and gated fusion with physiological output

EVALUATION & RESULTS



Ablation study — fusion (C5) significantly beats both single modalities

Video held-out macro-F1 (best fold: 0.6615)

0.6486

Fusion vs physio-only / video-only (Wilcoxon)

p < 0.001

- Full-frame input outperforms YOLO-cropped face input (0.70 vs 0.56 macro-F1) — a genuine empirical finding from this project.

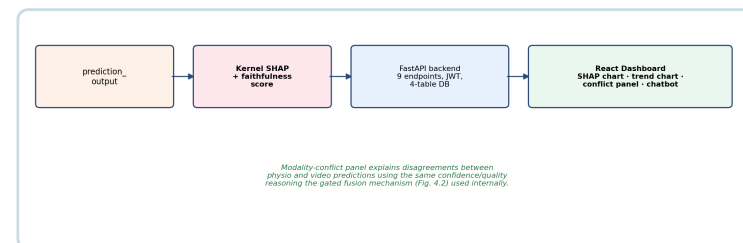
KEY FEATURES

- Actor-independent cross-validation (no actor overlap train/test).
- Entropy-based confidence × signal-quality penalty → L1-normalised fusion weights.
- Graceful degradation: falls back cleanly to a single modality when the other is missing/poor (verified by robustness tests).

MODULE 03 Explainability + Web Application

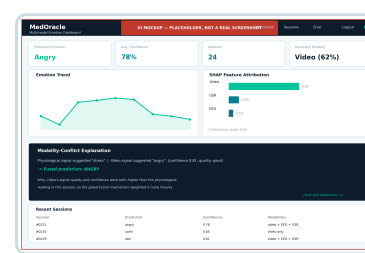
Kernel SHAP over the fused prediction, served through a FastAPI backend (9 endpoints, JWT auth) and a React dashboard with a novel modality-conflict explanation panel and an LLM-backed chatbot.

METHODOLOGY



Module 03 pipeline — SHAP explainability layer, FastAPI backend and React dashboard

EVALUATION & RESULTS



React dashboard layout

⚠ UI MOCKUP — not a real screenshot; to be replaced before printing

Automated tests passing (fusion, pipeline, harmonisation, HTTP)

82

REST API endpoints, real multimodal upload end-to-end

9

KEY FEATURES

- Kernel SHAP + faithfulness scoring (perturbation-based).
- Modality-conflict panel — explains physio/video disagreements using the fusion gate's own reasoning.
- Real file upload: video + EEG + GSR → fused, explained prediction.

Conclusion

MedOracle demonstrates a working, honestly-evaluated multimodal emotion recognition system. Its quality-aware gated fusion significantly outperforms any single modality and degrades gracefully when a sensor is unreliable — precisely the scenario a real deployment must handle. Where a component (the physiological module, under strict subject-independent evaluation) falls short, this project treats that as a rigorously diagnosed finding rather than a hidden weakness, and shows the system is explicitly designed to remain robust to it.

Key Results / Outcomes

- Video module generalises to unseen actors: **0.6486 macro-F1**, near-zero train/test gap.
- Multimodal fusion significantly outperforms either single modality (**p < 0.001**, Wilcoxon signed-rank).
- System degrades gracefully under missing/poor signal quality — verified, not assumed.
- Physiological module's honest limitation (**LOSO 0.179**) is diagnosed with evidence via four controlled experiments — not hidden.
- 82/82** automated tests passing; full working web app with real multimodal upload.